



- 3 A particle P is moving in a straight horizontal line. At time t s, the displacement of P from a fixed point O on the line is x m and the velocity of P is v ms⁻¹. The acceleration of P is $\frac{1}{2}(v^2 + 4)$ ms⁻² in the direction PO . Initially P is at O and is moving with velocity 2 ms⁻¹.

(a) Find an expression for x in terms of t .

[5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



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(b) Find the time when P next goes through O .

[2]

